

Topic Included: Number System & Representation (2021–2026)

Q1.

[CUET PG 2021 | 2's Complement Range]

The smallest integer that can be represented by an 8-bit number in 2's complement form is:

- (a) -256 (b) -128
(c) -127 (d) 0

Q2.

[CUET PG 2021 | 2's Complement Calculation]

2's complement of 1011011 is:

- (a) 0100011 (b) 0110101
(c) 0100111 (d) 0100101

Q3.

[CUET PG 2021 | Binary to Decimal]

The decimal equivalent of the binary number $(1011.01)_2$ is:

- (a) $(11.375)_{10}$ (b) $(10.123)_{10}$
(c) $(11.25)_{10}$ (d) $(9.23)_{10}$

Q4.

[CUET PG 2022 | Binary Multiplication]

The product of two four bit positive binary numbers 1011 and 0011 is:

- (a) 1000011 (b) 100001
(c) 100101 (d) 100111

Q5.

[CUET PG 2022 | Sign & Magnitude]

Representation of -11 in sign and magnitude is:

- (a) 11011 (b) 010111
(c) 10011 (d) 01110

Q6.

[CUET PG 2022 | 2's Complement Interpretation]

1001 is 2's complement representation of:

- (a) -7 (b) +9
(c) +6 (d) -6

Q7.

[CUET PG 2022 | Statements]

Statement I: 011010 is 2's complement of -37. Statement II: $111_{(8)}$ is octal of 73.

- (a) Both true (b) Both false
(c) I true, II false (d) I false, II true

Q8.

[CUET PG 2022 | Assertion & Reason]

Assertion A: Binary info represented in terms of 'zeros' & 'ones'. Reason R: Digital computers use voltages.

- (a) Both correct, R explains A (b) Both correct, R not explanation
(c) A correct, R incorrect (d) A incorrect, R correct

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Q9.

[CUET PG 2023 | 2's-Complement Adder]

Sign-magnitude numbers represent [---] numbers, and 2's complements stand for [---].

- (a) hexa, sign (b) sign, hexa
(c) positive, negative (d) negative, positive

Q10.

[CUET PG 2023 | Digital Electronics]

Consider statements regarding bases, Boolean SOP form, and D flip-flops for a mod(258) counter.

- (a) A only (b) A and B only
(c) A and C only (d) C only

Q11.

[CUET PG 2023 | Binary Numbers]

The number of 1's in the binary representation of $(3 \times 4096 + 15 \times 256 + 5 \times 16 + 3)$ are:

- (a) 8 (b) 29
(c) 10 (d) 12

Q12.

[CUET PG 2023 | IEEE-754]

Given 32-bit IEEE-754 format: 00111110011011010000000000000000. Closest decimal value?

- (a) 1.45×10^1 (b) 1.45×10^{-1}
(c) 2.27×10^{-1} (d) 2.27×10^1

Q13.

[CUET PG 2025 | 2's Complement Range]

The range of integers represented by 2's complement number system with n bits is:

- (a) -2^{n-1} to $2^{n-1}-1$ (b) $-(2^{n-1}-1)$ to $(2^{n-1}-1)$
(c) -2^{n-1} to 2^{n-1} (d) $-(2^{n-1}+1)$ to $(2^{n-1}-1)$

Q14.

[CUET PG 2025 | Radix System]

Given $\sqrt{(224)}_r = (13)_r$, where r is the radix. The value of r is:

- (a) 10 (b) 8
(c) 5 (d) 6

Q15.

[CUET PG 2025 | IEEE-754 Zero]

Which statement is true about IEEE-754 representation of +0 and -0?

- (a) Same sign bit, different mantissa bits. (b) Differ only in sign bit; exponent and mantissa are identical.
(c) Same representation because they are mathematically equivalent. (d) Differ in both exponent and mantissa bits.

Q16.

[CUET PG 2026/25 | Signed Complement]

Perform arithmetic addition of decimal numbers in List-I using signed-complement and match List-II.

List-I:	List-II:
(A) +6, +13	(I) 00000111
(B) -6, +13	(II) 00010011
(C) +6, -13	(III) 11101101
(D) -6, -13	(IV) 11111001

- (a) (A)-(II), (B)-(I), (C)-(III), (D)-(IV) (b) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
(c) (A)-(I), (B)-(II), (C)-(IV), (D)-(III) (d) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

ANSWER KEY (2021–2026)

Q. No.	Answer	Q. No.	Answer
1	b	9	c
2	d	10	b
3	c	11	d
4	b	12	c
5	a	13	a
6	a	14	c
7	d	15	b
8	a	16	b